F26-003 FYP-1 Mid-1

Investify

Educational equity decisions with evidence and checked explanations

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Working draft prepared 13 October 2026

Problem and research contribution

A decision label leaves important questions unanswered.

- Which observed features contributed to the result?
- Does the explanation agree with the recorded numbers?
- Did the evaluation use information available at that time?

FinBERT, XGBoost and TreeSHAP supply existing methods.
Investify adds temporal controls and a checked explanation path.
Predictive usefulness and user understanding require separate evidence.

Evidence: `report/chapters/chapter2_research.tex;deliverables/PROBLEM_AND_CONTRIBUTION.md`

Scope and acceptance checks

Current prototype

- Historical NVDA cases
- Five-session horizon
- 5 technical and 6 news features
- Separate risk comparison

Observable acceptance

- Validate the selected record
- Expose feature meaning and units
- Reject contradictory prose
- Verify the fallback separately

The social study remains separate. Phi-3-mini adaptation is later work.

Evidence: `report/chapters/chapter4_requirements.tex;code/tests/test_presenter_ui.py`

Mid-1 deliverables

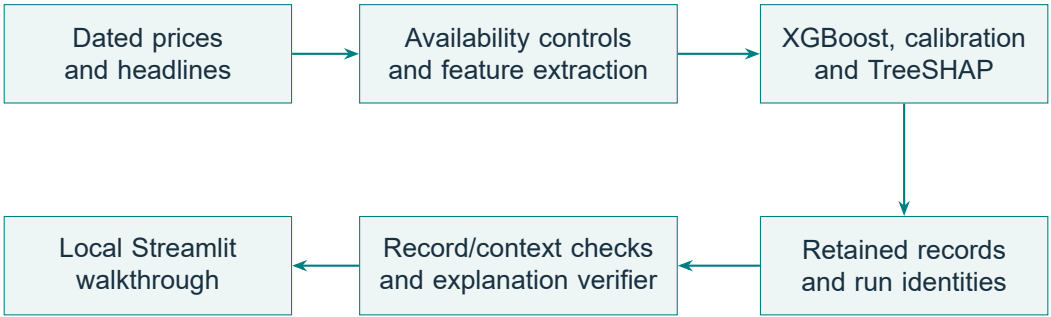
Week 8 formative supervisor checkpoint Recommended duration: 15 minutes

Required area	Prepared evidence
Report v1.0	Vision, related work, SRS, use cases, test plan, DEI and ethics
Appropriate design	Domain, components, layers, DFD, sequence, state and schema
Working iteration	Four-step demo, failure case, code walkthrough and tests
Maintenance	CI configuration, deployment diagram and setup manual

Genuine team contributions, baseline review and applicable approvals remain open.

Evidence: `sources/BS-FYP-Handbook-2026_DRAFT.pdfpp.11,27,37-38;planning/HANDBOOK_MID1_AUDIT.md`

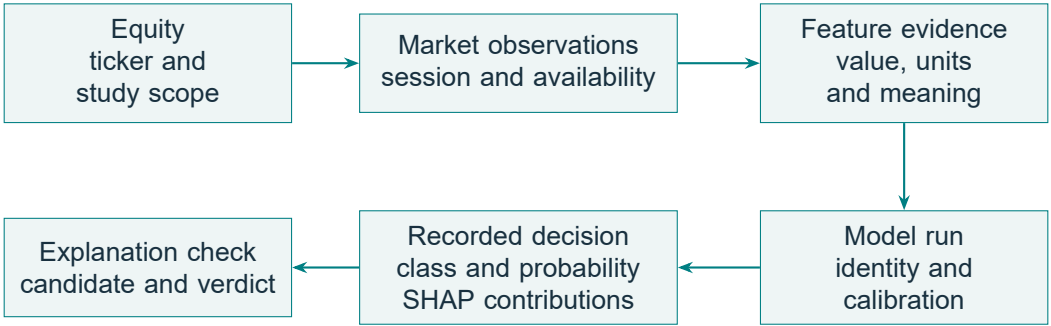
System architecture



Offline research creates the retained artifacts. The live demo reads them and checks explanations.

Evidence: `report/figures/mid1_pipeline.tex;code/app/presenter_data.py`

Domain model and file schema

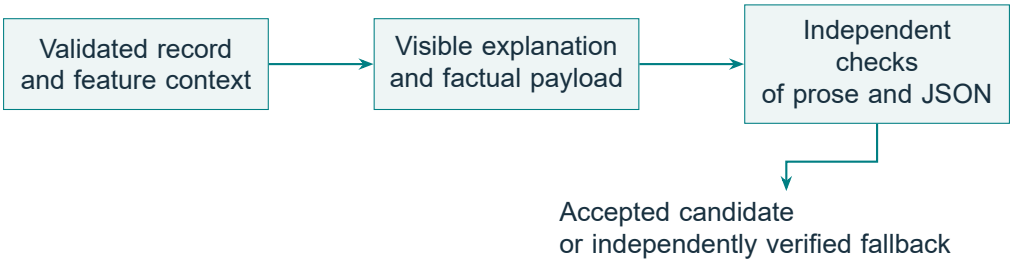


File schemas bind cases to source/session and model identity.

Current storage uses JSON/CSV artifacts. A SQL database is unnecessary here.

Evidence: `report/figures/mid1_equity_domain.tex;report/figures/mid1_schema.tex`

Explanation check and recovery



Demonstrable defect

Correct JSON beside a wrong visible confidence must fail.
The UI retains the candidate rejection and the separate fallback verdict.

Evidence: `code/src/investify/explain/verifier.py;code/tests/test_presenter_ui.py`

Live demonstration

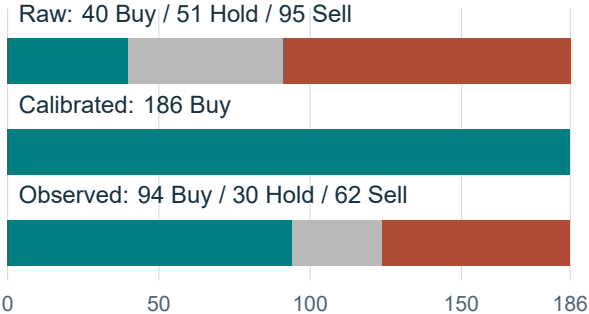
1. **Decision** Recorded class, probabilities and contributions
2. **Evidence** One feature and its definition at a time
3. **Risk profile** Explicit fictional profiles, separate comparison
4. **Explanation check** Wrong probability, rejection and fallback

`http://127.0.0.1:8507`

Default case: Fresh Alpaca frozen, NVDA, 21 December 2023.

Evidence: `report/chapters/chapter7_ui_manual.tex;evidence/presenter_calibration_ui_tests_20261013_run02.xml`

Calibration and decision collapse



Frozen classifier

Multiclass Brier
0.7929 to 0.6105

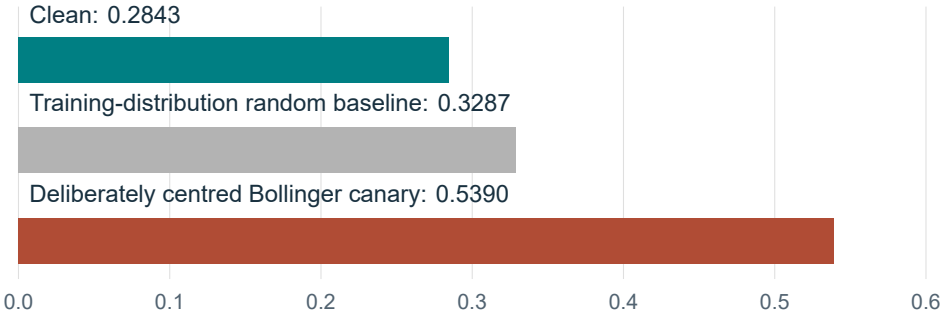
Calibration-prior comparator
0.6095

Evaluation: 186 sessions
Mar–Dec 2023

Saved map bounds force Buy for every raw probability vector.
Lower Brier alone does not establish useful classification.

Evidence: `evidence/fresh_alpaca_science_replay_20261012_run01.json;evidence/calibration_global_bounds_20261013.json`

Temporal leakage experiment



Five-session macro-F1, mean across six folds.

Clean lift: -0.0444 95% fold-bootstrap CI $[-0.0855, -0.0121]$.

Evidence: `evidence/fresh_alpaca_science_summary_20261012_run01.json`

Sentiment and teacher studies

Original sentiment candidates

Same 24 frozen MSFT AI references

Candidate	Macro-F1	Exact
CryptoBERT	0.2564	15/24
Cardiff	0.6016	16/24

Cardiff leads this benchmark.
Only two references are negative.
Final deployment selection is open.

Assistant teacher pilot

Six actual prose examples
Four training / two validation

The verifier accepts all six.
All six decisions are Buy.

Phi-3-mini remains untrained.
Human understanding is unmeasured.

Evidence: `evidence/social_study_review_20261013.json;deliverables/ASSISTANT_TEACHER_PILOT.md`

Equal work allocation and next steps

One-third responsibility each

Ahmed Javaid
Point-in-time harness, leakage tests, dual timestamps, parity test, deployment

Rafay Ul Haq
Risk profiling, sentiment pipeline, degraded operation, acceptance, frontend

Soban Ahmad
Decision engine, calibration, SHAP, verifier, later QLoRA distillation

Remaining review

Projector check and rehearsal

Member code defence

Metadata and baseline review

Social/risk criteria

Remote CI and deployment

Four slides and five minutes per member, including live demonstration.

Evidence: `sources/InvestifyUpdated(1).pptx, slide13; deliverables/WORK_DIVISION.md`